

LAB 05

PH142 Fall 2026

Announcements

- **Lab05:** due SUN 10/4 at 11:59 AM noon (extended)
- **Quiz04:** due SAT 10/3 at 11:59 PM midnight (extended)
- **MIDTERM I:** FRI 10/2 from 8am-9am



Week 6 Lecture Review

Independence of Events

Independent Events

- A and B are independent if knowing whether A occurs does not change the probability of B

Dependent Events

- A and B are dependent if knowing whether A occurs does change the probability of B

Week 6 Lecture Review

Screening Tests: Key Terms

Sensitivity: Probability the test is positive given disease present

- Sensitivity = $P(\text{Test+} \mid \text{Disease+})$

Specificity: Probability the test is negative given disease absent

- Specificity = $P(\text{Test-} \mid \text{Disease-})$

Sensitivity and specificity are properties of the test. They do not change based on prevalence of disease in the population.

Week 6 Lecture Review

Predictive Values

Positive Predictive Value (PPV): Probability the disease is present given a positive test result

- $PPV = P(\text{Disease+} \mid \text{Test+})$

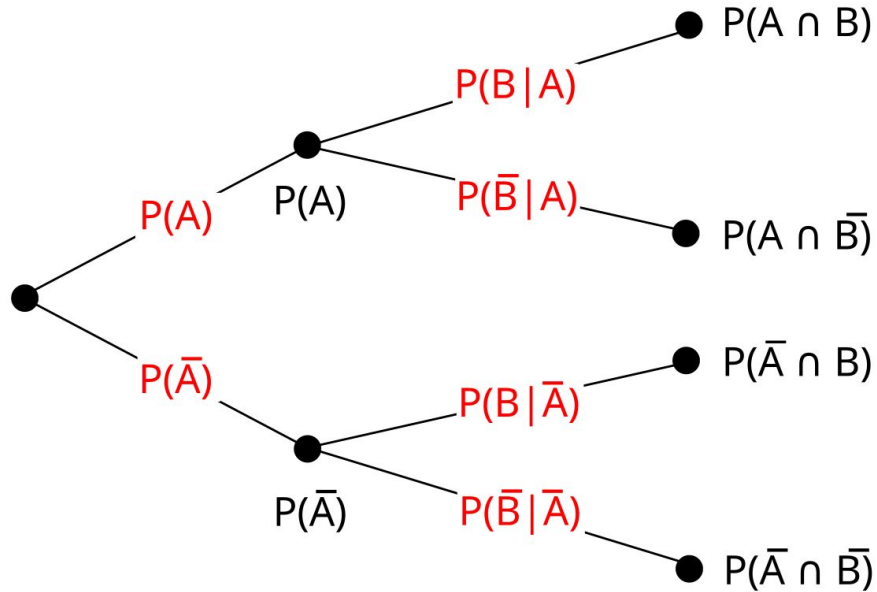
Negative Predictive Value (NPV): Probability the disease is absent given a negative test result

- $NPV = P(\text{Disease-} \mid \text{Test-})$

PPV and NPV change with the prevalence of disease in the population.

Week 6 Lecture Review

Tree Diagrams



Week 6 Lecture Review

The Normal Distribution

The normal distribution is a bell-shaped, symmetric distribution, with notation $\mathbf{X} \sim \mathbf{N}(\mu, \sigma)$

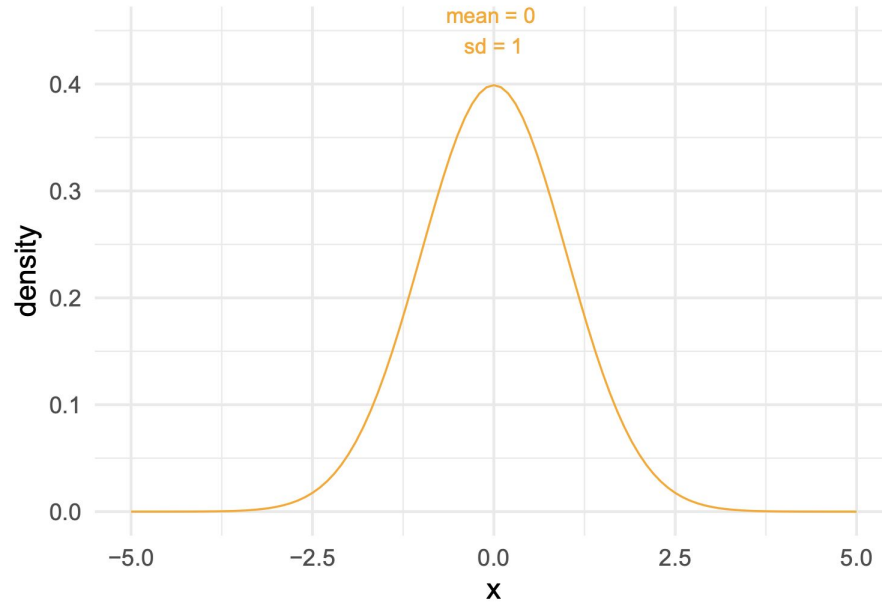
- μ = mean
- σ = standard deviation

R Functions:

- `pnorm()` → outputs the probability of value x or below
- `rnorm()` → generates random draws from the distribution

Week 6 Lecture Review

The Normal Distribution



Week 6 Lecture Review

Z-Scores

Z-score formula: $z = \frac{x - \mu}{\sigma}$

x = data point

μ = mean

σ = standard deviation

The Z-score tells you how many standard deviations the data point is from the mean.



LAB 05 Walkthrough

Lab Submission

- Follow the directions on the LAB05 file
- Submit using the **Terminal Tab** (next to the console in the bottom left pane)
 - Copy and paste the given line into the terminal
 - Follow prompts (NOTE: the terminal will **not** show your password being typed out!)
- **CHECK IN GRADESCOPE THAT ALL YOUR TESTS PASSED**